$$\int_0^{\pi/2} \sin^3 x \cos^{5/2} x\,dx \qquad m = 3, \; n = 5/2$$

$$\frac{\Gamma\frac{m+1}{2}\,\Gamma\frac{n+1}{2}}{2\,\Gamma\frac{m+n+2}{2}}$$

MTS 316 TEST. Time Allowed: 45 minutes. Answer ALL

1. Prove that $\beta(m,n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}}\,dx, \quad m, n > 0$, hence evaluate $\int_0^{\frac{\pi}{2}} \sin^{3} x \cos^{\frac{5}{2}} x\,dx$.

2. Use integration by parts to evaluate $\int (1+x)\sin(nx)\,dx$

3. An Engineer is interested in filtering all unwanted frequencies for a signal represented by an odd function

$f(x) = 1 + x, \; 0 < x < \pi$

$f(x + 2\pi) = f(x).$

Obtain its Fourier Sine Series.

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE, NIGERIA
DEPARTMENT OF MATHEMATICAL SCIENCES

EXAMINATION FOR THE DEGREE OF BACHELOR OF TECHNOLOGY (B.TECH)
COURSE TITLE: ENGINEERING MATHEMATICS II
COURSE CODE: MTS 316 — **NO OF UNITS: 3**
SEMESTER/SESSION: SECOND/2023/2024 — **TIME ALLOWED: 2hrs.**
INSTRUCTION: ANSWER ANY THREE QUESTIONS.

$\Gamma n = \int$

1. (a) Define Gamma function and Beta function. (2)

   (b) Evaluate the following integrals

   (i) $\int_0^{\infty} e^{-x^2} x^{2n-1} dx, \;\; n>1,$ (4) (ii) $\int_0^{\infty} e^{-\sqrt{x}} x^{\frac{1}{4}} dx$ (4) (14) (iii) $\int_0^{\infty} \frac{x^a}{a^x} dx.$ (4)

   (c) Prove that $\Gamma n \Gamma(1-n) = \dfrac{\pi}{\sin n\pi}$, (4) (5) hence evaluate $\int_0^{\frac{\pi}{2}} \sin^{10} x dx$. (4)

$\Gamma n =$ $\frac{\Gamma m \Gamma n}{\Gamma(m+n)} = \frac{\Gamma n \Gamma 1-n}{\Gamma(a+1-n)} = \frac{\Gamma n \Gamma 1-n}{\Gamma 1} = \frac{\Gamma n \Gamma 1-n}{1}$

2. (a) The Periodic signal of an AC Waveform is represented as

$$f(x) = \begin{cases} -x, & -\pi < x < 0 \\ 0, & 0 < x < \pi \end{cases}$$

$$f(x+2\pi) = f(x)$$

$\frac{\sin^n}{\cos^{n-1}} dx.$ $\ln \cos^{n-1} x.$ $\cos^{n-1} x = 0$ $\int_0^{\pi} \sin^n x \cos^{1-n} x \, dx = 0.$ $n = \pi$

As an Electric Engineer interested in the mitigation of the effects of higher-order harmonics in order to improve the performance and the efficiency of the electrical equipment, (a) Provide the graphical illustration of the periodic function that is not sinusoidal. (b) Considering the fact that $\cos(n\pi) = 1$ for n is even and $\cos(n\pi) = -1$ for n is odd, obtain the Fourier series.

(b) Given that

$$u_t(x,t) = u_{xx}(x,t) + \pi^2 e^{-24\pi^2 t} \sin 5\pi x, \;\; t>0, \;\; 0<x,<1$$

$$u(x,t) = 0, \;\; u(1,t) = 0, \;\; t>0,$$

$$f(x) = 3\sin 4\pi x, \;\; 0<x<1$$

(i) Classify the given equation according to linearity, homogeneity, and type of differential equation.

(ii) Identify the boundary and initial conditions of the given equation.

(iii) Use the method of eigenfunction expansion to solve the IBVP.

$\frac{\partial u}{\partial t} = \frac{\partial u^2}{\partial t^2} + \pi^2 e^{-24\pi^2 t} \sin 5\pi x.$ $\pi \sin^{-1} n\pi \Rightarrow \int_0^{\pi} x \sin^{-1} nx.$

$\frac{X'(x)}{X(x)} = \frac{T''(t)}{T(t)} = -\lambda.$

$\lambda = \frac{n^2\pi^2}{l^2}$

$\Gamma n \, \Gamma(1-n)$

$\Gamma n = \Gamma(n-1)_1$ $\Gamma n = \Gamma(n+1)-1)$

$\Gamma(1-n) = \Gamma((1-n)+1)-1) = \Gamma(1-n)$

3. (a) given a Sturm-Liouville system;

$$\frac{d}{dx}\left[p(x)\frac{dy}{dx}\right]+[q(x)+\lambda r(x)]y=0$$ (5)

$$a_1y(a)+a_2y'(a)=0,\ b_1y(b)+b_2y'(b)=0,\quad a\le x\le b$$

where $a_1, a_2, b_1, b_2, p(x), q(x), r(x), \lambda$ retain their usual definitions. Show that the eigenfunctions belonging to two different eigenvalues are orthogonal with respect to $r(x)$ in (a,b).

(b) Consider the Legendre's equations;

$$\left(1-x^2\right)P_m''(x)-2xP_m'(x)+m(m+1)P_m(x)=0 \qquad (1)$$

$$\left(1-x^2\right)P_n''(x)-2xP_n'(x)+n(n+1)P_n(x)=0 \qquad (2), \text{ where } m\ne n$$

Prove that the Legendre's Polynomials $P_m(x)$ and $P_n(x)$ are orthogonal in $-1<x<1$.

4. Solve the heat flow equation

$U_t=3U_{xx},\ 0\le x\le\pi,\ t>0$, when

$U(0,t)=U(\pi,t)=0$ for $t>0$ and

$U(x,0)=\sin 4x+3\sin 6x-\sin 10x$.

5. (a) Classify the general equation $a\dfrac{\partial^2U}{\partial x^2}+b\dfrac{\partial^2U}{\partial x\partial y}+c\dfrac{\partial^2U}{\partial y^2}=dU+e$ into the three primary classes of partial differential equation with an example with an example in each.

(b) Find the solution $U=U(x,y)$ of the linear partial differential equation $2x\dfrac{\partial U}{\partial x}+2y\dfrac{\partial U}{\partial y}=\alpha U+7$ where $x(0,s)=s,\ y(0,s)=1,\ U(0,s)=\varphi(s)$ and s is a parameter.

$$\int_x \frac{2x\partial u}{\partial x}+\int_x 2y\frac{\partial u}{\partial y}=\int_x \alpha u+7$$

$$x^2+2yx\frac{\partial u}{\partial y}+g(y)=\alpha u+7.$$